THE UNIVERSITY OF MICHIGAN
MICHIGAN SOCIETY OF FELLOWS
Rackham Building
Ann Arbor, Michigan 48109
(313) 763-1259
Richard Mandeberg

ADDENDUM TO THE GEN LOCK

In its initial design, the Gen Lock misplaces the leading edge of vertical blanking by one half line during the odd field. This circuit addition will fix that problem, and reestablish proper interlace to composite blanking. Vertical sync information is unchanged by this addition, as sync was already correctly interlaced in the previous design.

This addendum is useful to people who anticipate that much of their work will be time base corrected, or that they will be working often in a broadcast environment. The earlier design will work fine with all non-broadcast equipment, and its use does not preclude the ability to time base correct your videotapes.

NOTE: The addition of this circuit makes the construction of the Gen Lock more complex and difficult. If you don't understand how to put it in, use the original design. The module will work fine, and you can add the additional circuit later.

CONSTRUCTION

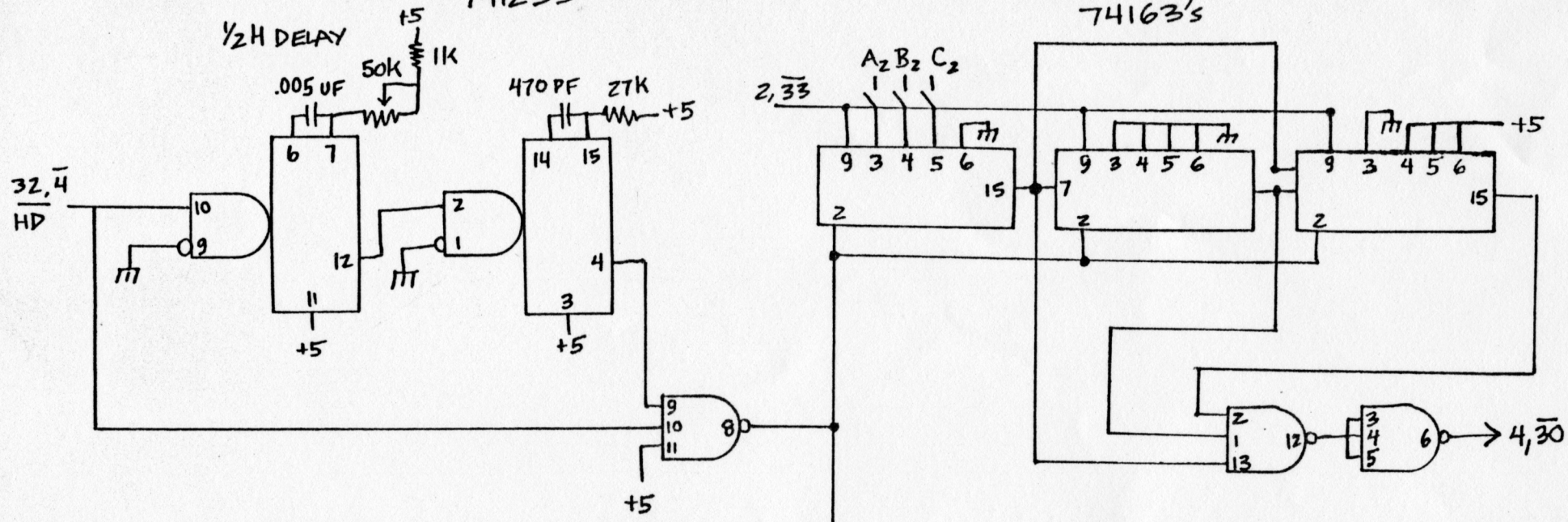
Build the circuit on perf board, wire wrapping the connections. Mount it above the digital board. Remember to bus ground and +5 volts to the board, and bypass +5 to ground with a .1uf cap. every few IC's. Mount the 74123 and the 50k trimmer near the rear of board, with the trimmer adjust facing back. Drill a hole in the back panel in the appropriate place for the 1/2 H adjustment. On the main digital board remove the 74163 which is in line with the 74123's, and remove the 74163 below it also. Do not wire wrap wires associated with these two IC's.

SET UP

Attach an oscilloscope probe to pin 8 of the 7410. Display it, and a high quality video signal (also inputted to the Gen Lock) on both channels of the scope. Trigger the scope to show the beginning of the vertical blanking interval, including equalization pulses and sync. Adjusting the 50k trimmer will change the phase of the 1/2 line pulse. Using every other equalization pulse as a guide, adjust these pulses so that their leading edge is coincident, or slightly ahead of the leading edge of the equalization pulses.

74123's

½H DELAY



74163's

74123 PIN 16. +5
PIN 8. m

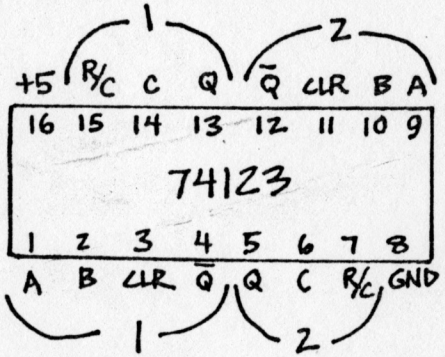
74163 PIN 16. +5
PIN 8. m
PIN 1. +5
ALL UNUSED PINS 7, 10. +5

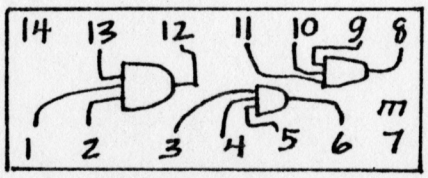
7410 PIN 14. +5
PIN 7. m

ON DIGITAL BOARD MODIFY TO:

+5 22, $\overline{21}$, 32, $\overline{21}$, 31, $\overline{21}$

GND 34, $\overline{21}$, 33, $\overline{21}$





7410

GEN·LOCK MODIFICATION 3/79

INPUT

The input module suppresses the sync and clamps the signal coming from the camera, thus preparing any standard video input for the IP.

The video signal is inputted at JI1 and composite sync (-4v.) is inputted at JI2.

There are three electrical modules in the chassis box, so replicate work three times. There is only one JI2 in the chassis box and its terminal should be connected to the same spot on all three cards. Remember to buss (connect) +12 and -12 and ground from center card to top and bottom card. Also buss +12 and -12 to front panel for R1(s) and R2(s); take from center card.

TEST STUFF:

R1 is adjusted untill most of the sync is suppressed but blacks are not clipped. (An osscilloscope is necessary.) R2 is adjusted until the video signal is symetrical, about 0 volts and has a magnitude of approximately 1 volt peak-to-peak into a 75 ohm load.

The output is available at JO1, JO2, JO3 and JO4.

NOTE: these adjustments may have to retuned, off and on, for different video sources.

SYNC GENERATOR

This module generates full NTSC color sync conforming to RS 170 EIA after trimming the master osscillator. The sync generator should stay well within the broadcast standard.

All outputs are -4v. into 75 ohm except the 14 MHZ. (JO13) which is an open collector TTL. JO13 is not used except for work with digital computers.

JI1 horizontal reset and JI2 vertical reset are not implemented in full.

The sync generator requires starting pulses which are provided by capacitors associated with JI1 and JI2. Time must be allotted after power-down before power-up (at least 30 seconds) to assure proper starting of sync generator.

Master osscillator trim:
Adjust the master osscillator frequency to 14.318180 MHZ. A convenient way to do this is to run a lead from horizontal drive and wrap it (still fully insulated) around the antennae of a TV receiver. Tune the receiver to a station on feed from a major network (in color). You will notice two vertical lines or one vertical bar drifting across the screen. Adjust the master osscillator with a long insulated screwdriver through the access hole in the front panel of module until vertical lines do not drift.

A-D56

10K Ten Turn Trimmers

COLOR ENCODER

This module generates a N.T.S.C. compatible color video signal. The front panel switch selects between an internal color bar generator, and the front panel Red, Green, and Blue inputs. Its inputs and outputs are;

JI1 3.58 MHz Color Carrier (1-2 volts p-p) sub CARRIER
JI2 Burst Flag
JI3 Composite Sync
JI4 Horizontal Drive (not used)
JI5 Composite Blanking
JI6 Vertical Drive (not used)
JI7 Red (+.5 to -.5 volts)
JI8 Green " "
JI9 Blue " "

JO1 N.T.S.C. video
JO2 " "
JO3 " "

The Luminance Board combines Red, Green, and Blue to form the luminance component of the video signal. The inverted (-Y) form is used within the Color Encoder, and sent to the R-Y, B-Y, and Chroma Modulator Boards. Blanking and sync are also inserted on the Luminance Board.

The R-Y, and B-Y Boards accept red, and blue respectively, and form the color difference signals. Each signal is low pass filtered to help bandwidth limit the resultant chroma signal. Burst Flag is used to effect proper burst height and hue adjustment.

The Chroma Modulator Board generates chroma from the color difference signals, and combines it with the luminance component. The Bar Generator Board serves two functions. Controlled by the front panel switch, it generates the proper red, green, and blue signals for making color bars. Composite blanking is also converted from an EIA type pulse to a TTL level signal and sent to other boards within the module.

Mount the Bar Generator, and B-Y Boards in the top of the module. R-Y, and Chroma Modulator Boards go on the bottom, with the Luminance Board in the middle.

Carrier balance and white balance may drift during warm up. Make all internal adjustments after the IP has come up to temperature. In a broadcast environment, a proc. amp. may be necessary to prevent accidental over chroma modulation.

Richard Mandeberg 6/80

SET-UP

Set-up should be done only after the IP has warmed up for at least twenty to thirty minutes. You will need to know the proper gain and amplitude levels (either in I.R.E. units or voltages) for N.T.S.C. video. A vectorscope, if accessible, may be substituted in some of the steps.

1. Attach all inputs except for red, green, blue, and 3.58. Display one or two lines of output on the oscilloscope. Set all internal trimmers to center position.

2. Set the front panel switch in the bars position. Adjust RT10 and RT11 for the proper descending stairstep associated with the luminance component of color bars. Adjust RT1 (Y Gain), R4 (Pedestal), and R8 (Sync Height) for luminance amplitude, pedestal, and sync.

3. Attach the 3.58 input. Adjust RT2 and RT5 for minimum carrier leakage during blanking.

4. Turn R7, the front panel Chroma Gain to the full gain position. Switch SW to video. Adjust RT3 and RT6 for minimum carrier leakage during active video.

5. Switch back to bars. Adjust RT4 and RT7 for white balance.

6. Adjustment of the B-Y Gain (RT8) requires repeated tweaking of white balance and carrier leakage. RT8 is adjusted to give the proper chroma amplitude relationships between the different colors of color bars. Alternatively, it may be set by looking at the direct outputs of the R-Y, and B-Y Boards, and setting them for the correct gain relationship. Tweak RT8, and repeat steps 4, 5, and 6 until correct. Remember, RT3 and RT6 are adjusted with SW in the video mode.

7. With R7 still at full gain, adjust RT9 for 100% chroma.

8. Adjust R6 for burst height. Using a properly tuned monitor set R5 for the correct hue or phase of color bars.

9. In the video mode, with the encoder driven to white, the white balance may need slight adjustment.

PARTS

All parts are standard IP parts except;

LM1889N	Chroma Modulator	National
DM74LS14N	Hex Schmitt Inverter	National
DM74LS74N	Dual Flip Flop	National
LM555CN	Timer	National

MOD Chroma Modulator P.C. Board from E.C.I.

When # of moduals changes readjustment is necessary only on RT2, RT3, RT5, & RT6 (Carrier Balance & Carrier Null)

3

J11
3.58

J12
BURST FLG

J17
RED

R1 100

J13
COMPSYNC

J18
GRN

R2 100

J14
HORIZ DR.

J19
BLUE

R3 100

FRONT FACE

J15
COMP BLK

J16
VERT DR.

R4 10K
RED

SW R
SW G
SW B
BARS/VIDEO

COLOR ENCODER

R5 10K
HUE

R4 PED.
R5 HUE

R6 10K
BURST HEIGHT

J01
OUT

R8
SYNC. HEIGHT

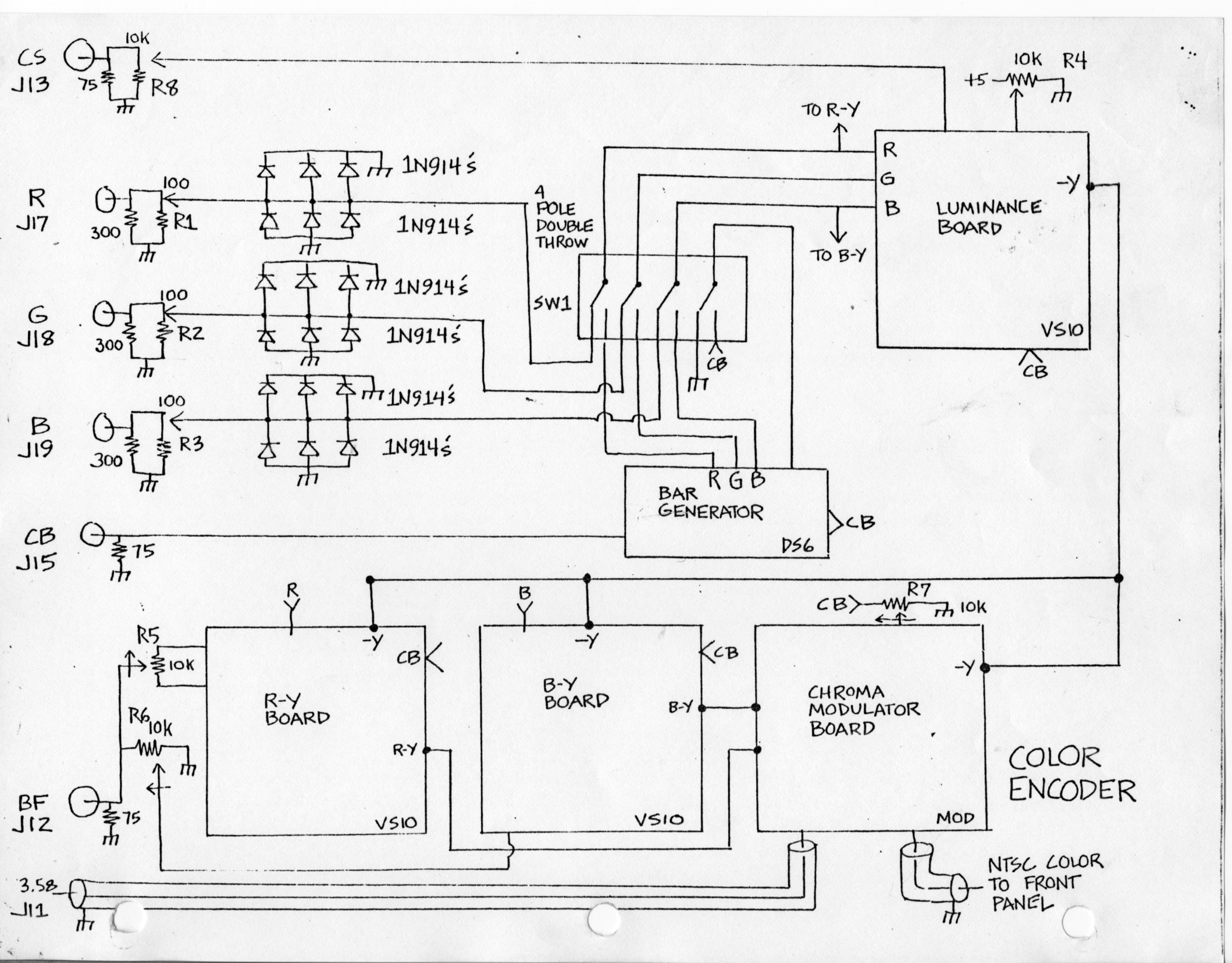
R7 10K
CHROMA GAIN

R7 CHROMA

R8 10K
SYNC HEIGHT

J02
OUT

J03
OUT


CS
J13
10K
75
R8
10K R4
+5
TO R-Y
TO B-Y
R
G
B
LUMINANCE
BOARD
-Y
VS10
CB
R
J17
100
300
R1
1N914's
4
POLE
DOUBLE
THROW
SW1
G
J18
R2
B
J19
R3
BAR
GENERATOR
R G B
DS6
CB
J15
R5
10K
R6
10K
BF
J12
R-Y
BOARD
B-Y
BOARD
CHROMA
MODULATOR
BOARD
MOD
R7
COLOR
ENCODER
3.58
J11
NTSC COLOR
TO FRONT
PANEL